

Domain Controller and Active Directory Lab

Part 8 – Organizational Units(OU), Users, Account Global Domain Local Permissions (AGDLP) using PowerShell

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Technologies Used

- **Oracle VirtualBox**
- **Windows Server 2022**
- **Server Manager**
- **Active Directory Domain Services**
- **PowerShell**

OBJECTIVE:

The objective of this lab is to create multiple Organizational Units (OU) for different departments with users assigned to them utilizing PowerShell. Once the OUs are created, Account Global Group Domain Local Group Permission (AGDLP) procedure will be implemented, ensuring users assigned to departments have appropriate permissions and access.

Lab Setup:

This lab continues from Part 7 where Users accounts were created, grouped and assigned their respective permissions pertaining to their group using the AD Graphics User Interface (GUI).

For this lab PowerShell will be utilized to create group accounts in OU. PowerShell streamlines creation and organization process by minimizing click-through

```
PS C:\Windows\system32> New-ADUser -Name "Sarah Connor" -SamAccountName "SConnor" -Path "OU=Human Resources,DC=corp,DC=corson,DC=com" -UserPrincipalName "SConnor@corp.corson.com" -AccountPassword (Read-Host -AsSecureString "Enter Password") -Enabled $true
Enter Password: *****
```

Figure 1: PowerShell is utilized to create the user account Sarah Connor using the New-ADUser cmdlet. The command assigns the user's name, sAMAccountName, User Principal Name (UPN), password, and account status. The -Path parameter places the newly created account directly into the Human Resources OU.

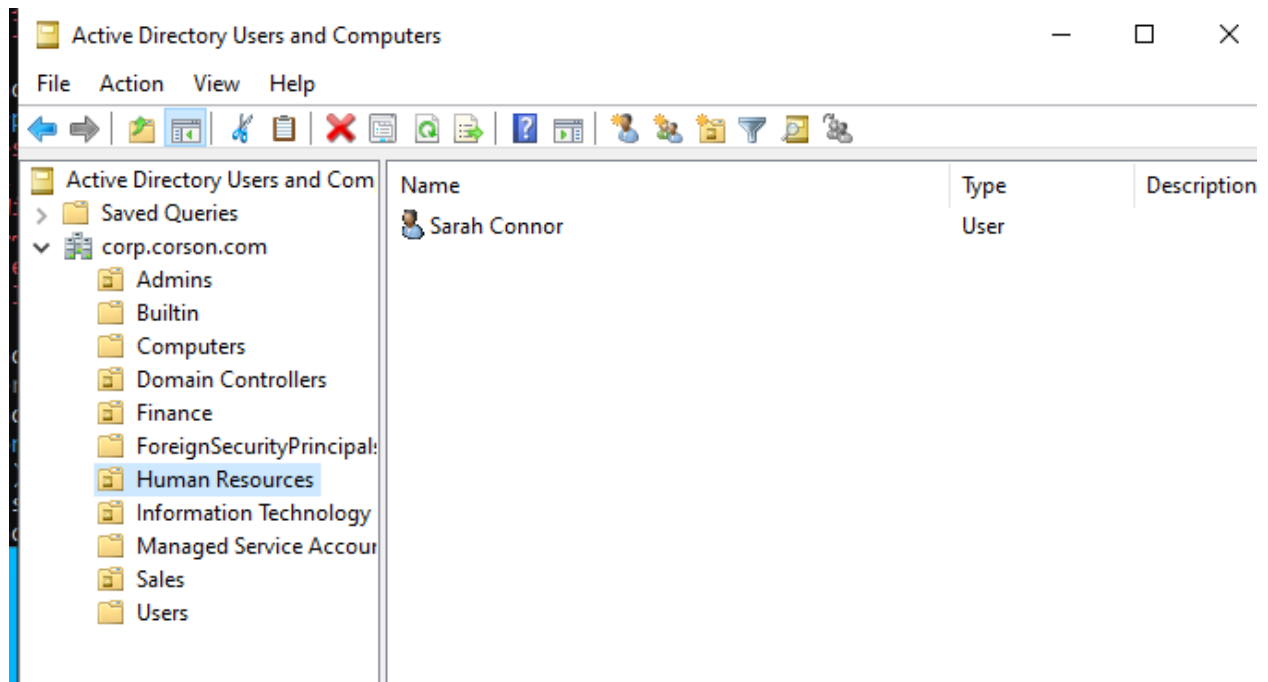


Figure 2: Active Directory Users and Computers verifies that the Sarah Connor user account was successfully created and placed within the Human Resources OU using PowerShell.

```
PS C:\Windows\system32> New-ADGroup -Path "OU=Human Resources, DC=corp,DC=corson,DC=com" -Name "GG_Human_Resources_Users" -GroupScope Global
PS C:\Windows\system32>
```

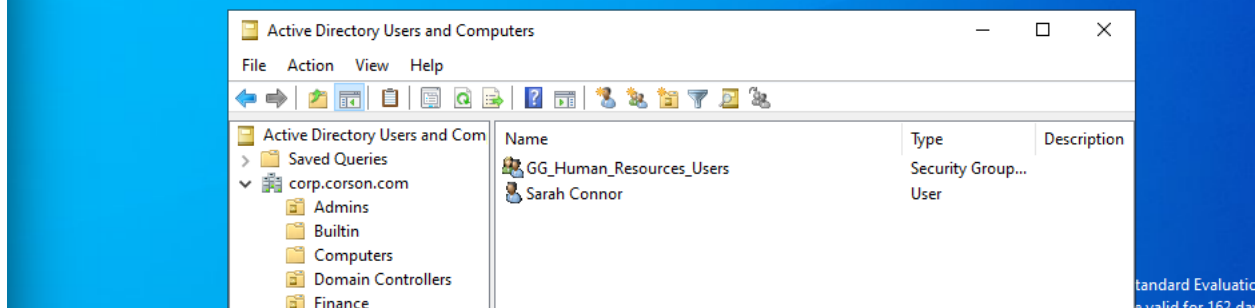


Figure 3: PowerShell is utilized to create the GG_Human_Resources_Users Global Security Group within the Human Resources OU. This group will contain user accounts belonging to the Human Resources department and represents the Global Group portion of the AGDLP structure.

```
PS C:\Windows\system32> Add-ADGroupMember -Identity GG_Human_Resources_Users -Members 'CN=Sarah Connor,OU=Human Resources,DC=corp,DC=Corson,DC=com'
PS C:\Windows\system32>
```

Figure 4: The Add-ADGroupMember PowerShell cmdlet is utilized to add Sarah Connor to the GG_Human_Resources_Users Global Security Group, completing the Account to Global Group portion of AGDLP.

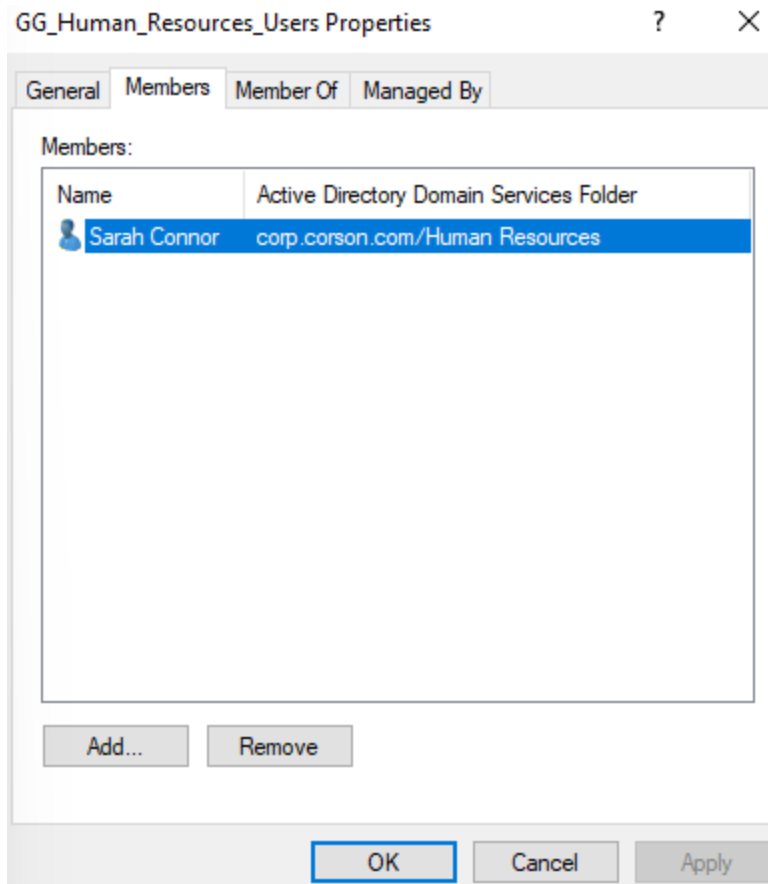


Figure 5: The Members tab of GG_Human_Resources_Users verifies that Sarah Connor was successfully added as a member of the Global Security Group.

```
PS C:\Windows\system32> New-ADGroup -Path "OU=Human Resources, DC=corp,DC=corson,DC=com" -Name "DL_Human_Resources_Files_Modify" -GroupScope DomainLocal
```

Figure 6: PowerShell is utilized to create the DL_Human_Resources_Files_Modify Domain Local Security Group within the Human Resources OU. This group will be used to assign permissions to Human Resources resources rather than assigning permissions directly to individual users.

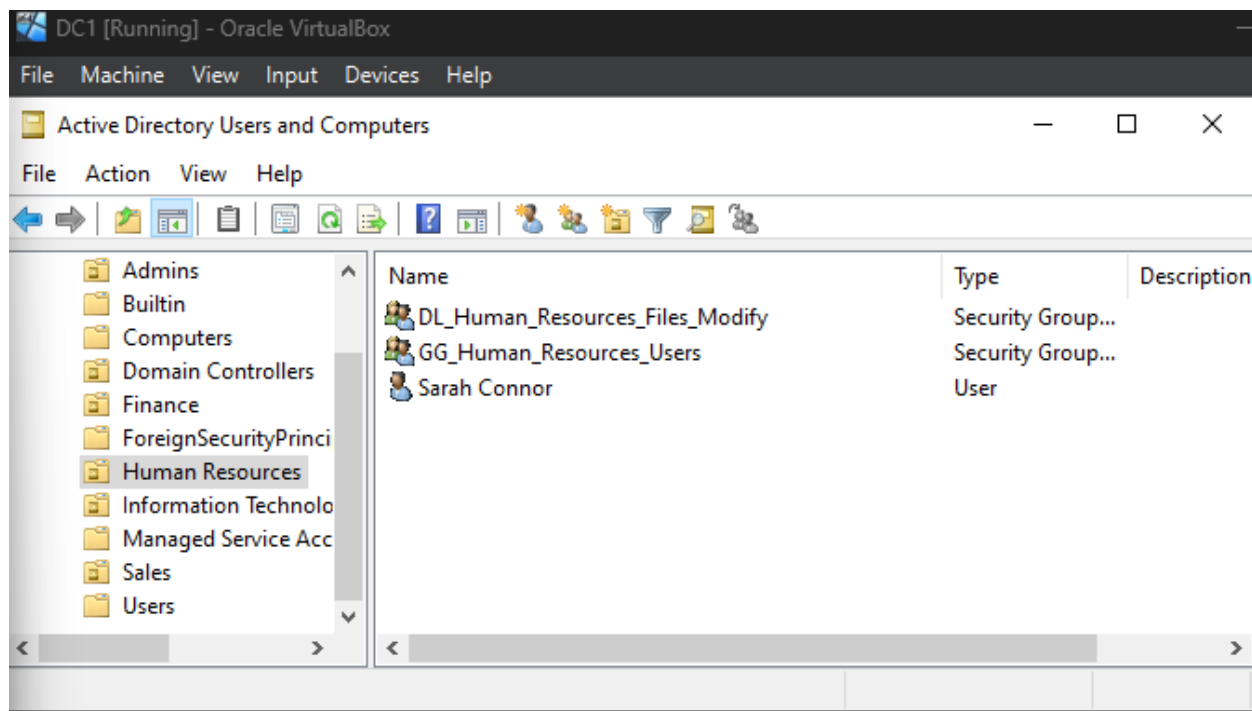


Figure 7: Active Directory Users and Computers verifies the creation of the GG_Human_Resources_Users Global Security Group and DL_Human_Resources_Files_Modify Domain Local Security Group within the Human Resources OU.

```
PS C:\Windows\system32> Add-ADGroupMember -Identity DL_Human_Resources_Files_Modify -Members 'CN=GG_Human_Resources_Users,OU=Human Resources,DC=corp,DC=corson,DC=com'
PS C:\Windows\system32>
```

Figure 8: The GG_Human_Resources_Users Global Security Group is added as a member of the DL_Human_Resources_Files_Modify Domain Local Security Group using the Add-ADGroupMember cmdlet. This establishes the Global-to-Domain Local portion of AGDLP.

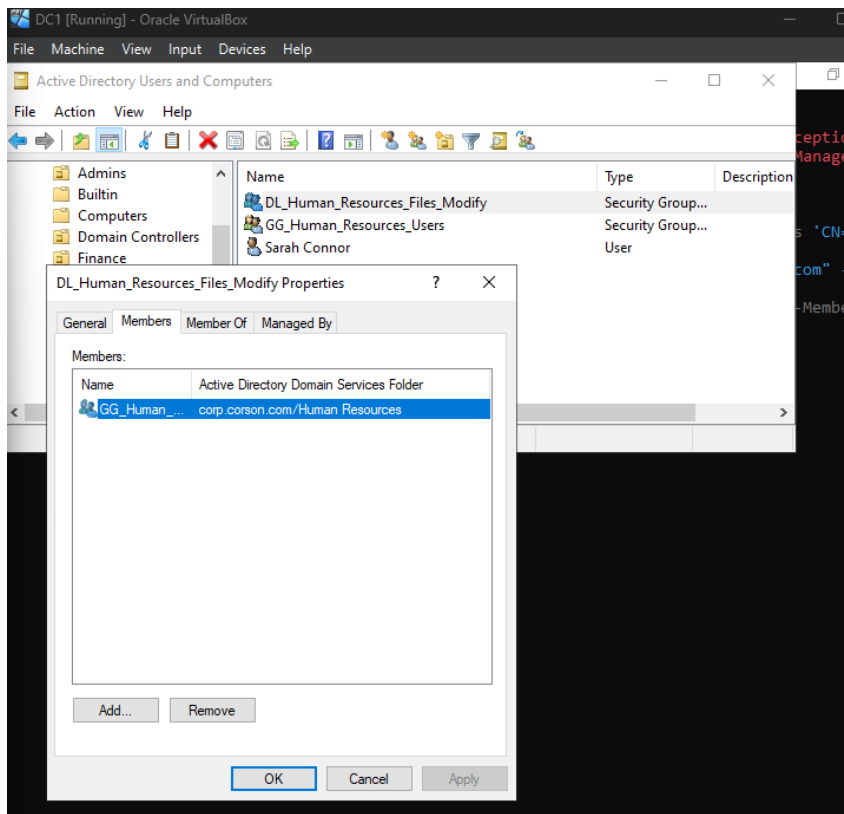


Figure 9: Active Directory Users and Computers verifies that GG_Human_Resources_Users is successfully nested within the DL_Human_Resources_Files_Modify Domain Local Security Group.

```
PS C:\Windows\system32> New-Item -Path "C:\Shares\Human Resources" -ItemType Directory

Directory: C:\Shares

Mode                LastWriteTime         Length Name
----                -
d-----          8/22/2026   3:47 PM              Human Resources

PS C:\Windows\system32>
```

Figure 10: The New-Item PowerShell cmdlet is utilized to create the C:\Shares\Human Resources directory. This directory will serve as the shared departmental resource for Human Resources users.

```
PS C:\Windows\system32> New-SmbShare -Name "Human Resources" -Path "C:\Shares\Human Resources"

Name                ScopeName Path                Description
-----
Human Resources *    C:\Shares\Human Resources
```

Figure 11: The New-SmbShare PowerShell cmdlet is utilized to create the Human Resources network share and associate it with the C:\Shares\Human Resources directory.

```
PS C:\Windows\system32> Grant-SmbShareAccess -Name "Human Resources" -AccountName "DL_Human_Resources_Files_Modify" -AccessRight Full

Confirm
Are you sure you want to perform this action?
Performing operation 'Modify' on Target '*,Human Resources'.
[Y] Yes [A] Yes to All [N] No [L] No to All [S] Suspend [?] Help (default is "Y"): Y

Name                ScopeName AccountName                AccessControlType AccessRight
-----
Human Resources *    Everyone                Allow                Read
Human Resources *    CORP\DL_Human_Resources_Files_Modify Allow                Full
```

Figure 12: The Grant-SmbShareAccess cmdlet grants the DL_Human_Resources_Files_Modify Domain Local Security Group Full access at the SMB share level. NTFS permissions will provide more restrictive control over what members can perform within the directory.

```
PS C:\Windows\system32> $human_resource_acl = Get-Acl -Path "C:\Shares\Human Resources"
```

Figure 13: The existing NTFS Access Control List (ACL) for the Human Resources folder is retrieved using Get-Acl. The ACL displays the existing users and groups that currently have permissions to the directory, along with their access and inheritance settings.

```
PS C:\Windows\system32> $mdRule = New-Object System.Security.AccessControl.FileSystemAccessRule("CORP\DL_Human_Resources_Files_Modify","Modify","ContainerInherit,ObjectInherit","None","Allow")
PS C:\Windows\system32> _
```

Figure 14: A new FileSystemAccessRule is created for the DL_Human_Resources_Files_Modify group. The rule assigns Modify permission and configures the permission to apply to the Human Resources folder, its subfolders, and files.

```
PS C:\Windows\system32> $human_resource_acl.SetAccessRule($mdRule)
PS C:\Windows\system32> _
```

Figure 15: The newly created Modify access rule is added to the existing Human Resources ACL using SetAccessRule. The updated ACL now contains an explicit permission entry for CORP\DL_Human_Resources_Files_Modify.

```
PS C:\Windows\system32> $mdRule = New-Object System.Security.AccessControl.FileSystemAccessRule("CORP\DL_Human_Resources_Files_Modify","Modify","ContainerInherit,ObjectInherit","None","Allow")
PS C:\Windows\system32> $human_resource_acl.SetAccessRule($mdRule)
PS C:\Windows\system32> $human_resource_acl.Access

FileSystemRights : Modify, Synchronize
AccessControlType : Allow
IdentityReference : CORP\DL_Human_Resources_Files_Modify
IsInherited : False
InheritanceFlags : ContainerInherit, ObjectInherit
PropagationFlags : None

FileSystemRights : FullControl
AccessControlType : Allow
IdentityReference : NT AUTHORITY\SYSTEM
IsInherited : True
InheritanceFlags : ContainerInherit, ObjectInherit
PropagationFlags : None

FileSystemRights : FullControl
AccessControlType : Allow
IdentityReference : BUILTIN\Administrators
IsInherited : True
InheritanceFlags : ContainerInherit, ObjectInherit
PropagationFlags : None

FileSystemRights : ReadAndExecute, Synchronize
AccessControlType : Allow
IdentityReference : BUILTIN\Users
IsInherited : True
InheritanceFlags : ContainerInherit, ObjectInherit
PropagationFlags : None
```

Figure 16: The Set-Acl cmdlet applies the modified ACL to C:\Shares\Human Resources. Get-Acl is then utilized to retrieve the permissions directly from the folder and verify that DL_Human_Resources_Files_Modify successfully received Modify permission.

What Changed?

Before this lab, Active Directory users, groups, group memberships, shared folders, and permissions were primarily configured through the Active Directory and Windows graphical user interfaces.

PowerShell was utilized to perform the same administrative tasks through the command line. A Human Resources user account was created and placed directly into the appropriate Organizational Unit. A Global Security Group was then created for Human Resources users, and the user account was assigned as a member.

A Domain Local Security Group was created and the Human Resources Global Group was nested within it following the AGDLP model. A departmental Human Resources folder and SMB network share were then created using PowerShell. Share level access was granted to the Domain Local group, while NTFS Modify permissions were applied to the Human Resources directory, subfolders, and files.